

The ab-set: proofs

July 7, 2026

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This file proves the description of $K(a, b)$ stated in `ab_set.md`. Notation is as there: $O = (0, 0)$, $B = (b, 0)$, $A = a u(120^\circ)$, $u(\theta) = (\cos \theta, \sin \theta)$, $u^\perp(\theta) = u(\theta + 90^\circ)$, $h = \frac{\sqrt{3}}{2}$, $\sigma = a + b$, $R^2 = a^2 + ab + b^2$, $C = \{ru(\theta) : r \geq 0, \theta \in [0^\circ, 120^\circ]\}$,

$$U = \bigcup \{T : T \text{ unit equilateral, } \{A, O, B\} \subset T\}, \quad K = U \cap C.$$

For a compact $S \subset \mathbb{R}^2$ write $g_S(\theta) = \max_{p \in S} \langle p, u(\theta) \rangle$ (support function), and for the three-point configuration $g(\theta) := g_{\{A, O, B\}}(\theta) = \max\{0, a \cos(\theta - 120^\circ), b \cos \theta\}$.

1. The fitting criterion

Every unit equilateral triangle with outer unit normals $n_k = u(\varphi + 120^\circ k)$, $k = 0, 1, 2$, has the form

$$T(\varphi; c_0, c_1, c_2) = \{x : \langle x, n_k \rangle \leq c_k, \ k = 0, 1, 2\}, \quad c_0 + c_1 + c_2 = h. \quad (1.1)$$

Lemma 1.1. (i) For $c_0 + c_1 + c_2 \geq 0$ the set (1.1) is an equilateral triangle of side $\frac{2}{\sqrt{3}}(c_0 + c_1 + c_2)$; in particular it is a unit triangle iff $c_0 + c_1 + c_2 = h$, and every unit triangle with these normals arises this way. (ii) A compact S is contained in some unit triangle with normals $n_k(\varphi)$ iff

$$F_S(\varphi) := \sum_{k=0}^2 g_S(\varphi + 120^\circ k) \leq h. \quad (1.2)$$

Proof. (i) Since $n_0 + n_1 + n_2 = 0$, the intersection of the three half-planes is invariant in shape: translating so that the incenter (the point x^* with $\langle x^*, n_k \rangle = c_k - \rho$ for all k , i.e. equidistant ρ from the three lines) sits at the origin, we get the equilateral triangle with inradius $\rho = \frac{c_0 + c_1 + c_2}{3}$; a unit equilateral triangle has inradius $\frac{1}{2\sqrt{3}} = \frac{h}{3}$. Equivalently this is Viviani's theorem: for any point x in a unit equilateral triangle, the distances to the three sides sum to the height h ; and $\sum_k \langle x, n_k \rangle = 0$. (ii) $S \subset T(\varphi; c)$ iff $c_k \geq g_S(\varphi + 120^\circ k)$ for each k ; such c with $\sum c_k = h$ exists iff (1.2). $\square$

Corollary 1.2. $x \in U$ iff $\min_\varphi F_{\{A,O,B,x\}}(\varphi) \leq h$. Moreover, writing $F(\varphi) := F_{\{A,O,B\}}(\varphi)$ and $\Phi = \{\varphi : F(\varphi) \leq h\}$ (period 120°),

$$U = \bigcup_{\varphi \in \Phi} U_\varphi, \quad U_\varphi := \{x : F_{\{A,O,B,x\}}(\varphi) \leq h\}. \quad (1.3)$$

Lemma 1.3 (the hexagon U_φ). Let $m_k = g(\varphi + 120^\circ k)$ and $s = h - F(\varphi)$ (≥ 0 for $\varphi \in \Phi$). Then

$$U_\varphi = \left\{ x : \langle x, n_k \rangle \leq m_k + s \quad (k=0,1,2), \quad \langle x, -n_k \rangle \leq h - m_k \quad (k=0,1,2) \right\}. \quad (1.4)$$

Proof. $F_{\{A,O,B,x\}}(\varphi) = \sum_k \max(m_k, \langle x, n_k \rangle) \leq h = \sum_k m_k + s$. Expanding the sum of maxima by which terms exceed m_k , and using $\sum_k \langle x, n_k \rangle = 0$ for the triple term, this is equivalent to: for every k , $\langle x, n_k \rangle - m_k \leq s$; and for every pair $i \neq j$, $(\langle x, n_i \rangle - m_i) + (\langle x, n_j \rangle - m_j) \leq s$, i.e. $\langle x, -n_k \rangle \leq m_i + m_j + s = h - m_k$ for $\{i, j, k\} = \{0, 1, 2\}$. $\square$

Geometrically U_φ is the union of all unit triangles of orientation φ containing $\{A, O, B\}$: a hexagon with alternating outer sides (normals n_k) and inner sides (normals $-n_k$); its outer/inner corner in direction “between n_i and n_j ” is the free vertex of the triangle whose other two sides touch the supports.

2. The three-point data

Lemma 2.1 (supports). Let $\theta_{AB} \in [30^\circ, 90^\circ]$ be defined by $\cos \theta_{AB} = \frac{\sqrt{3}a}{2R}$, $\sin \theta_{AB} = \frac{a+2b}{2R}$ (any value for $R = 0$); $u(\theta_{AB})$ is the outer normal of the segment AB , and the distance from O to the line AB is $d_{AB} = \frac{\sqrt{3}ab}{2R}$. Then the maximizer in $g(\theta)$ is B for $\theta \in (-90^\circ, \theta_{AB})$, A for $\theta \in (\theta_{AB}, 210^\circ)$, and O for $\theta \in (210^\circ, 270^\circ)$, with ties at the three transition angles. (*Direct verification from $g = \max(0, a \cos(\theta - 120^\circ), b \cos \theta)$.*)

Lemma 2.2 (the function F). On a period, with all angles mod 120° :

$$F(\varphi) = \begin{cases} \sigma \cos \varphi, & \varphi \in [-30^\circ, 30^\circ], \\ R \cos(\varphi - \theta_{AB} + 30^\circ), & \varphi \in [30^\circ, \theta_{AB}], \\ R \cos(\varphi - \theta_{AB} - 30^\circ), & \varphi \in [\theta_{AB}, 90^\circ]. \end{cases} \quad (2.1)$$

Hence $\min F = \frac{\sqrt{3}}{2}R$ (at $\varphi = \theta_{AB}$), the two local maxima are $F(0^\circ) = \sigma$ and $F(\theta_{AB} \pm 30^\circ) = R$ (+ if $a > b$, - if $b > a$; for $a = b$ they merge), and the secondary local minimum is $F(30^\circ) = F(90^\circ) = \frac{\sqrt{3}}{2}\sigma$ (attained at 90° if $a > b$, at 30° if $b > a$).

Proof. For $\varphi \in (0^\circ, 30^\circ)$ the three normals $\varphi, \varphi + 120^\circ, \varphi + 240^\circ$ have supports B, A, O (Lemma 2.1), so $F = b \cos \varphi + a \cos \varphi + 0 = \sigma \cos \varphi$; the case $\varphi \in (90^\circ, 120^\circ)$ is the mirror image. For $\varphi \in (30^\circ, 90^\circ)$ the supports are $(\max(A, B), A, B)$ giving $F = \max(a \cos(\varphi - 120^\circ), b \cos \varphi) + a \cos \varphi + b \cos(\varphi - 120^\circ)$; using $\cos \varphi + \cos(\varphi - 120^\circ) = \cos(\varphi - 60^\circ)$ and writing $a \cos \varphi + b \cos(\varphi - 60^\circ) = R \cos(\varphi - \delta)$ with $\delta = \theta_{AB} - 30^\circ$ (a direct check of $R \cos \delta = a + \frac{b}{2}$, $R \sin \delta = \frac{\sqrt{3}}{2}b$), one gets $F = \max(R \cos(\varphi - \delta), R \cos(\varphi - \delta - 60^\circ))$, whose two branches swap exactly at $\varphi = \delta + 30^\circ = \theta_{AB}$. Values and monotonicity follow, using $\frac{\sqrt{3}}{2}\sigma \leq R \leq \sigma$ (the first is $(a - b)^2 \geq 0$, the second is $ab \geq 0$). $\square$

Corollary 2.3 (existence; bands). $U \neq \emptyset$ iff $R \leq 1$; at $R = 1$, U is the single unit triangle with base $[A, B]$ and third vertex on the O -side of the line AB (the minimal enclosing triangle at orientation θ_{AB}). For $R < 1$, comparing h with the critical values $\frac{\sqrt{3}}{2}R \leq \frac{\sqrt{3}}{2}\sigma \leq R \leq \sigma$ yields exactly the four band structures of Φ stated in the bands table of `ab_set.md`, with endpoints:

- outer gap ($\sigma > \frac{\sqrt{3}}{2}$): Φ misses the interval $(-\psi_1, \psi_1)$ around $\varphi \equiv 0$, where $\cos \psi_1 = \frac{\sqrt{3}}{2\sigma}$; this uses the first branch of (2.1), which requires $\psi_1 \leq 30^\circ$, i.e. $\sigma \leq 1$ (for $\sigma > 1$ see the merged case below);
- mid gap ($R > \frac{\sqrt{3}}{2}$): the gap around $\varphi = \theta_{AB} \pm 30^\circ$ is $(\theta_{AB} \pm 30^\circ - \kappa, \theta_{AB} \pm 30^\circ + \kappa)$ with $\cos \kappa = \frac{\sqrt{3}}{2R}$ (+ for $a > b$);
- for $\sigma > 1$ the two gaps merge and $\Phi = [\theta_{AB} - 30^\circ + \kappa, \theta_{AB} + 30^\circ - \kappa]$.

3. Global structure of U and the radial function

Lemma 3.1. U is compact, $U \subseteq \bar{B}(P, 1)$ for each $P \in \{A, O, B\}$, and U is star-shaped with respect to every point of $\text{conv}\{A, O, B\}$. Consequently $K = U \cap C$ is compact and star-shaped w.r.t. O , and for $\theta \in (0^\circ, 120^\circ)$

$$K \cap \{ru(\theta) : r \geq 0\} = \{ru(\theta) : 0 \leq r \leq r^*(\theta)\}, \quad r^*(\theta) = \max\{r : ru(\theta) \in U\}.$$

Proof. Boundedness and the ball containments: any two points of a unit equilateral triangle are at distance ≤ 1 (its diameter). Compactness: the family of admissible triangles is compact (orientations in the compact set Φ , translations bounded), so the union is closed. Star-shape: if $x \in T \supseteq \{A, O, B\}$ then $T \supseteq [y, x]$ for every $y \in \text{conv}\{A, O, B\} \subseteq T$ by convexity. $\square$

For fixed θ and $\varphi \in \Phi$ let

$$\rho_\varphi(\theta) = \max\{r \geq 0 : ru(\theta) \in U_\varphi\}$$

(well-defined: $O \in U_\varphi$, U_φ compact convex). By (1.3),

$$r^*(\theta) = \max_{\varphi \in \Phi} \rho_\varphi(\theta). \quad (3.1)$$

By Lemma 1.3, $\rho_\varphi(\theta)$ is computed from the six linear constraints; equivalently, with $c_k(\varphi) = \cos(\theta - \varphi - 120^\circ k)$,

$$\rho_\varphi(\theta) = \max \left\{ r : \sum_k \max(m_k(\varphi), r c_k(\varphi)) \leq h \right\},$$

a strictly increasing piecewise-linear equation in r (its exact solution formula is what `rho_of` in `verify_ab_set.py` implements).

4. Classification of boundary configurations

Fix $\theta \in (0^\circ, 120^\circ)$ and let $r = r^*(\theta)$, $x = ru(\theta) \in \partial U$, and let $\varphi^* \in \Phi$ attain the max in (3.1). Write T^* for the *minimal* triangle at orientation φ^* containing $\{A, O, B, x\}$ (sides at the support values); by maximality $F_{\{A, O, B, x\}}(\varphi^*) = h$, so T^* is a unit triangle, $x \in \partial T^*$, and each side of T^* touches at least one of A, O, B, x .

Lemma 4.1 (first-order conditions). One of the following holds at φ^* :

(E) $\varphi^* \in \partial \Phi$ (i.e. $F(\varphi^*) = h$): then x lies on a side of the extreme triangle T_{φ^*} , on the side line through the 3-point support of that side.

(S) $\varphi^* \in \text{int } \Phi$ and near φ^* the support points of all three sides are locally unique and constant. Then $\frac{d}{d\varphi} F_{\{A, O, B, x\}}(\varphi^*) = 0$, i.e. $\sum_k \langle P_k, u^\perp(\varphi^* + 120^\circ k) \rangle = 0$ where P_k is the support point of side k . This forces (writing $R_{\pm 120}$ for rotations):

- (S1) x on one side, say side 0, other supports $Q_1, Q_2 \in \{A, O, B\}$: then $x = -R_{-120^\circ} Q_1 - R_{120^\circ} Q_2 + h u(\varphi^*)$: x lies on the circle of radius h about one of the seven centers

$$O, A, B, -a u(0^\circ), -b u(120^\circ), -(a u(0^\circ) + b u(120^\circ)), \sigma u(60^\circ).$$

- (S2) x at a vertex (on two sides), the third side supported by $Q \in \{A, O, B\}$: then $x - Q \parallel$ the third normal and $\langle Q - x, n \rangle = h$, i.e. $|x - Q| = h$: circle of radius h about Q .

(T) $\varphi^* \in \text{int } \Phi$ and φ^* is a kink of $F_{\{A, O, B, x\}}$: two points tie as support of some side at φ^* , and the one-sided derivatives straddle 0. The possible ties:

- (T1) x ties with $Q \in \{A, O, B\}$ on a side, and x supports only that side. Then $F_{\{A, O, B, x\}}(\varphi^*) = F(\varphi^*) = h$, so in fact $\varphi^* \in \partial \Phi$ — this is (E) again, with x on the extreme-triangle side through Q .
- (T2) x ties with Q on one side and x also supports a second side (x at a vertex v of T^* , Q on a side through v), third side supported by $Q_2 \in \{A, O, B\}$: a one-parameter family; eliminating the triangle parameters gives the quartic E_{QQ_2} (computed in §6). For $Q_2 = Q$ it degenerates to the circle $|x - Q| = 1$.
- (T3) two of $\{A, O, B\}$ tie on a side **not** containing x . The tie pins that side's normal: $u(210^\circ)$ for $\{O, A\}$ (side flush on the line OA), $u(270^\circ)$ for $\{O, B\}$ (flush on line OB), $u(\theta_{AB})$ for $\{A, B\}$. With the flush side fixed, either
 - (a) x is at the opposite vertex: lines L_A ($\langle x, u(30^\circ) \rangle = h$), L_B ($\langle x, u(90^\circ) \rangle = h$), or the AB -analogue $\langle x, u(\theta_{AB}) \rangle = d_{AB} - h$ (never in C , cf. §9); or
 - (b) x is on one of the two remaining sides and the last side is supported by a point or vertex: the flush-slide lines $M_A, M_B, \lambda_B, \lambda_A$ and the horizontal/slant variants through A or B (all derived in §6; only the four named ones are ever exposed, see §9).

Proof. $F_{\{A,O,B,x\}}$ is a sum of maxima of finitely many smooth sinusoids in φ , hence piecewise-smooth with kinks exactly where a support changes; at an interior minimum either the derivative vanishes (S) or the subdifferential interval $[\partial^-, \partial^+]$ contains 0 at a kink (T). The derivative of a support term with unique support point P is $\langle P, u^\perp \rangle$. In case (S1) the two displayed equations ($F = h$ and $F' = 0$) determine x 's components along $u(\varphi^*)$ and $u^\perp(\varphi^*)$; assembling them gives $x = h u(\varphi^*) - R_{-120^\circ} Q_1 - R_{120^\circ} Q_2$, and letting φ^* vary traces the stated circle; the seven centers are the values of $-R_{-120^\circ} Q_1 - R_{120^\circ} Q_2$ over the support assignments that are realizable for some direction (Lemma 2.1 excludes $(Q_1, Q_2) \in \{(O, A), (B, O)\}$). (S2), (T1)–(T3) are the analogous eliminations; (T2) is carried out in §6. Note every case lands in the **catalog**: extreme-triangle side lines (E/T1), seven h -circles (S1, S2), quartics $E_{Q_1 Q_2}$ and circles c_A, c_B plus $|x - O| = 1$ (T2), and flush lines (T3). $\square$

Proposition 4.2 (master formula). For each θ , $r^*(\theta)$ equals the maximum of the radial coordinates (in direction θ) over the finite catalog of Lemma 4.1 restricted to the configurations realizable at that direction. In particular ∂K minus the two cone edges is covered by finitely many explicit algebraic curves, namely the catalog of `ab_set.md` §1 (the h -circles and the extra flush lines are in the catalog but turn out never to be exposed inside C ; §9).

Proof. Combine (3.1) with Lemma 4.1: the maximizing φ^* exists by compactness of Φ and continuity, and each alternative places x on a catalog curve. $\square$

Lemma 4.3 (the circle $|x - O| = 1$ never appears). For $a, b > 0$ the circle $|x - O| = 1$ contains no boundary point of U : a configuration (T2) with $Q = Q_2 = O$ places O at a vertex of T^* , and $A, B \in T^*$ would lie in the 60° wedge of T^* at O — impossible since $\angle AOB = 120^\circ$. (For $a = 0$, when $A = O$, the circle does appear; it is then the common degeneration of c_A and E_{AO} , consistently with the case tables.) By contrast c_A and c_B are realizable: the wedge at the vertex A must contain O and B , which subtend an angle $\leq 60^\circ$ at A whenever $|AO|, |AB| \geq$ the relevant chords — quantitatively this is the containment computation of §5 and pins the exposed arcs exactly between the corners listed in the case tables. $\square$

5. Lower bounds = realizability

Lemma 5.1. Let $\Gamma \subset C$ be the closed chain claimed for a case in `ab_set.md` §4 (including the two cone-edge segments), and suppose every point of the chain lies in U . Then the compact region S enclosed by Γ satisfies $S \subseteq K$.

Proof. Each claimed chain is by construction the graph of a positive continuous function $\theta \mapsto r_\Gamma(\theta)$ over $[0^\circ, 120^\circ]$ (checked case by case: consecutive arcs meet at the listed corners with increasing θ), so $S = \{ru(\theta) : 0 \leq r \leq r_\Gamma(\theta)\}$. For $x = ru(\theta) \in S$ take the chain point $x_0 = r_\Gamma(\theta)u(\theta) \in U$; a triangle $T \ni x_0$ containing $\{A, O, B\}$ contains O , hence the segment $[O, x_0] \ni x$. Also $x \in C$. $\square$

So the lower bound reduces to: **every chain point is covered by an explicit triangle**. For each arc type we exhibit the triangle and the containment inequalities:

- **Ray segments.** For $x = (t, 0)$, $0 \leq t \leq t_{B^*} = \frac{\sqrt{4-3a^2}-a}{2}$: the four points O, A, B, x lie in a unit triangle iff the three-point set $\{O, A, (\max(t, b), 0)\}$ does (the fourth point lies in the convex hull); by Corollary 2.3 applied to parameters $(a, \max(t, b))$ this holds iff $a^2 + at + t^2 \leq 1$, i.e. $t \leq t_{B^*}$, which also shows $B^* \in c_A$. Mirror for the other ray.
- **c_A -arcs** (x with $|x - A| = 1$): take T with vertices x, A , and $v = A + R_{-60^\circ}(x - A)$ (the choice on the clockwise side). Containment of O and B amounts to four scalar inequalities,

e.g. $\langle O - A, n \rangle \leq 0$ etc., which hold exactly for the θ -range between the listed corners (endpoints of the range are exactly where O or B reaches a side, which is where the adjacent arc of the chain takes over; this containment is re-verified pointwise along every claimed arc in `verify_ab_set.py`).

- **L_A -segments:** T = the flush triangle with base $[x - hu(30^\circ) - \frac{1}{2}u(120^\circ), x - hu(30^\circ) + \frac{1}{2}u(120^\circ)]$ on the line OA and apex x ; containment of O, A = both in the base segment; containment of B : $\langle B, u(30^\circ) \rangle = \frac{\sqrt{3}}{2}b \leq h$ automatic, and the two lateral constraints hold exactly between the corners listed (P_1, G_1, P_2 -range). Similarly L_B , and M_A, M_B (the rigid triangles T_A^*, T_B^* ; e.g. $B \in T_A^*$ iff $\langle B, u(-30^\circ) \rangle \leq \frac{\sqrt{3}}{2}(1-a)$ iff $\sigma \leq 1$), and λ_B, λ_A (sliding flush triangles; the slide interval is nonempty exactly on the exposed sub-segment).
- **Extreme segments $s_\bullet$:** $T = T_{\varphi_e}$ itself.
- **Quartic arcs E_{Q_2} :** T = the defining pivot triangle of the configuration (T2), which by construction contains Q, Q_2, x ; the containment of the remaining point is a single scalar inequality along the arc, which holds precisely between the listed corners. (Equality at a corner is exactly the corner condition — e.g. the E_{BO} -arc pivots keep A inside until φ reaches the value at which A touches a side, which is the handover corner $W_\mp$ or G'_1 or P_2 ... as listed.)

All these scalar inequalities are polynomial in (a, b) and the arc parameter; they are verified symbolically/numerically in `verify_ab_set.py` over each case region (see `ab_set_verification.md`).

6. Derivation of the catalog curves

Below φ is the orientation, $n_k = u(\varphi + 120^\circ k)$; complex notation $z \simeq (X, Y)$.

6.1 Circles of radius h (S1/S2). Immediate from Lemma 4.1: e.g. for (S1) with $(Q_1, Q_2) = (A, B)$, the side line through x is $\langle y, u(\varphi) \rangle = h - \langle A, n_1 \rangle - \langle B, n_2 \rangle = h - \langle au(0^\circ) + bu(120^\circ), u(\varphi) \rangle$, whose envelope over φ is the circle of radius h centered $-(au(0^\circ) + bu(120^\circ))$. These circles never win the maximum (3.1) inside C (they are dominated by the (T2)/(T3)/(E) families; part of the §9 comparisons), so they do not appear in the case tables.

6.2 The quartics (T2). Configuration: x at the vertex $n_0 \wedge n_1$, companion Q on the side with normal n_1 (chirality; the other chirality exchanges the roles of n_0, n_1), Q_2 on the side with normal n_2 . Solving $\langle x, n_0 \rangle = h - \langle Q, n_1 \rangle - \langle Q_2, n_2 \rangle$, $\langle x, n_1 \rangle = \langle Q, n_1 \rangle$ for x gives (with $w = e^{i\varphi}$)

$$z(\varphi) = e^{i(\varphi+30^\circ)} + \frac{Qe^{i30^\circ} - Q_2e^{i150^\circ}}{\sqrt{3}} - \frac{i(\bar{Q} - \bar{Q}_2)}{\sqrt{3}}e^{2i\varphi} \quad (6.1)$$

(and the mirror-chirality formula with $e^{i(\varphi+90^\circ)}$, e^{-i30° , $+e^{-i30^\circ}(\bar{Q} - \bar{Q}_2)$). These are limaçons: a fixed circle of radius 1 plus a second harmonic of amplitude $\frac{|Q-Q_2|}{\sqrt{3}}$. Eliminating φ (resultants; done in `derive_quartics.py`) yields the implicit quartics $E_{AO}, E_{BO}, E_{BA}, E_{AB}$ printed in `ab_set.md` §1. For $Q = Q_2$, (6.1) is the circle $|z - Q| = 1$; this is how c_A ($Q = A$) and c_B ($Q = B$) arise, while $Q = O$ is excluded by Lemma 4.3. Smoothness on the exposed ranges: a limaçon $e^{it} + \gamma e^{2it}$ is regular iff $|\gamma| \neq \frac{1}{2}$; here $|\gamma| \in \{\frac{a}{\sqrt{3}}, \frac{b}{\sqrt{3}}, \frac{R}{\sqrt{3}}\}$ and on the exposed ranges $a < \frac{1}{2}$ (resp. $b < \frac{1}{2}$, $R < \frac{1}{2}$... in fact $\sigma < \frac{1}{2}$) holds, so $|\gamma| < \frac{1}{2}$ —no cusps or loops on exposed arcs.

6.3 Flush lines (T3). Side flush on line OA means normal $n_1 = u(210^\circ)$, support value 0; then $c_0 + c_2 = h$. If x is the opposite vertex, $\langle x, n_0 \rangle + \langle x, n_2 \rangle = \langle x, -n_1 \rangle = \langle x, u(30^\circ) \rangle = h$: the line L_A . If x is on the side with normal $u(330^\circ)$ (the “right” side) and the third side touches A at its end (vertex

at A), the triangle is the rigid T_A^* and x moves on $\langle x, u(330^\circ) \rangle = h - \langle A, u(90^\circ) \rangle = \frac{\sqrt{3}}{2}(1-a)$: the line M_A . If instead the third side touches B , the triangle slides and $\langle x, u(90^\circ) \rangle = h - \langle B, u(330^\circ) \rangle = \frac{\sqrt{3}}{2}(1-b)$: the line λ_B . The OB -flush family gives L_B , M_B , λ_A mirror-wise. (The remaining combinatorial options — third side touching O , or the AB -flush family — produce lines that are never exposed inside C ; §9.)

6.4 Extreme segments (E). At $\varphi_e \in \partial\Phi$, U_{φ_e} is the single triangle T_{φ_e} ; its side lines are $\langle x, n_k \rangle = m_k(\varphi_e)$ with the supports of Lemma 2.1. The values of φ_e per band (Corollary 2.3) give exactly the line formulas $s_A^{\text{out}}, s_B^{\text{out}}, s_A^{\text{mid}}, s_B^{\text{mid}}, s_A^{\text{IV}}, s_B^{\text{IV}}$ of `ab_set.md` §1 after substituting $\cos \psi_1 = \frac{\sqrt{3}}{2\sigma}$, $\sin \psi_1 = \frac{a}{2\sigma}$, $\cos \kappa = \frac{\sqrt{3}}{2R}$, $\sin \kappa = \frac{p}{2R}$ and the θ_{AB} data of Lemma 2.1 (routine trigonometry; verified in sympy).

7. Corner identities and thresholds

All statements below are polynomial identities in (a, b) , verified exactly in sympy (`derive_thresholds.py`; also re-checked numerically). Redundant spurious factors like $(\sigma \pm 1)$ powers are discarded against the band inequalities.

1. $P_1 = A + (1, 0)$ lies on c_A , L_A and E_{AO} for all a, b ; E_{AO} is tangent to L_A at P_1 iff $a = \frac{1}{2}$. For $a < \frac{1}{2}$ the arc of E_{AO} emanating from P_1 lies strictly above L_A (outside the half-plane $\langle x, u(30^\circ) \rangle \leq h$), for $a > \frac{1}{2}$ strictly below — whence the predicate α .
2. $P_3 = B + u(120^\circ)$ lies on $c_B \cap L_B \cap E_{BO}$; tangency of E_{BO}, L_B at P_3 iff $b = \frac{1}{2}$ — predicate β . (Mirror of 1.)
3. $P_2 = B + (1-b)u(60^\circ) \in L_A \cap E_{BA} \cap E_{BO}$, and E_{BA} is tangent to L_A at P_2 iff $\sigma = \frac{1}{2}$; $P'_2 = A + (1-a)u(60^\circ) \in L_B \cap E_{AB} \cap E_{AO}$, tangency of E_{AB}, L_B at P'_2 iff $\sigma = \frac{1}{2}$ — predicate ε . (E_{BO} is transversal to L_A at P_2 unless $b = \frac{1}{2}$; E_{AO} transversal to L_B at P'_2 unless $a = \frac{1}{2}$.)
4. Restrictions to the flush lines factor as

$$E_{AO}|_{L_A} \propto (t - (a - \frac{1}{2})) \mathcal{C}_A(t), \quad E_{BO}|_{L_B} \propto (t - (b - \frac{1}{2})) \mathcal{C}_B(t),$$

$$E_{BA}|_{L_A} \propto (t - (\frac{1}{2} - b)) \mathcal{D}_A(t), \quad E_{AB}|_{L_B} \propto (t - (\frac{1}{2} - a)) \mathcal{D}_B(t),$$

with the cubics of `ab_set.md` §2 (parametrizations: on L_A , $x = (\frac{3}{4} - \frac{t}{2}, \frac{\sqrt{3}}{4}(1+2t))$, so $t = a - \frac{1}{2} \leftrightarrow P_1$ and $t = \frac{1}{2} - b \leftrightarrow P_2$; on L_B , $x = (t, \frac{\sqrt{3}}{2})$, so $t = b - \frac{1}{2} \leftrightarrow P_3$, $t = \frac{1}{2} - a \leftrightarrow P'_2$). This identifies G_1, G'_1, G_2, G'_2 and shows they are the *unique* extra crossings in the relevant ranges.

5. E_{BO} passes through P'_2 iff $(\sigma - 1)\mathcal{W} = 0$ where $\mathcal{W} = a^3 + a^2b - a^2 + a - b$; equivalently $\mathcal{C}_B(\frac{1}{2}-a) = -8\mathcal{W}$. Note the exact identity $w_- = -\mathcal{W}$ (expand: $-\mathcal{W} = b(1-a^2) - a(a^2-a+1)$). Since the E_{BO} -arc through P_2 and P_3 is the outermost long arc on its side, the B -side flank (the E_{AB}/L_B pieces around P'_2) is exposed iff E_{BO} passes radially *inside* P'_2 , i.e. iff $w_- > 0$; the threshold is exactly $P'_2 \in E_{BO}$, i.e. $\mathcal{W} = 0$. Mirror: the A -side flank is exposed iff $w_+ > 0$, threshold $P_2 \in E_{AO}$, via $P_2 \in E_{AO} \iff (\sigma - 1)(ab^2 - a + b^3 - b^2 + b) = 0$ and $w_+ = -(ab^2 - a + b^3 - b^2 + b)$.
6. $\eta_\pm$ (band I, ε -regime): the segment of L_B between E_{AB} and E_{BO} appears exactly when their crossing Q'_2 passes below L_B ; at the threshold E_{AB}, E_{BO}, L_B are concurrent, i.e. the cubics $\mathcal{D}_B, \mathcal{C}_B$ have a common root: $\text{Res}_t(\mathcal{D}_B, \mathcal{C}_B) = 0$, the polynomial $\mathcal{E}(a, b)$ (sign convention checked on the region: exposure $\iff \mathcal{E} > 0$; numerically the transition was located independently by bisection at $(a, b) = (0.135, b)$, giving $b = 0.27225\dots$, at which the two real roots

coincide to $2 \cdot 10^{-16}$ and $\mathcal{E}$ vanishes to $6 \cdot 10^{-17}$). The event is independent of the state of the other side, so it also splits the both-flanks region into I.1/I.1a/I.1b/I.1c; e.g. at $(0.24, 0.255)$ both $\eta_{\pm}$ hold and the verified chain is the ten-arc I.1c chain. Mirror on L_A : $\mathcal{E}(b, a)$.

7. Outer-dip flush (λ_B): exposed iff the level $\frac{\sqrt{3}}{2}(1-b)$ exceeds the height of the crossing $s_A^{\text{out}} \cap s_B^{\text{out}}$; eliminating radicals gives the polynomial threshold $\mathcal{G}(a, b)$ of `ab_set.md` §3 (factor of degree 5; the discarded factors $(\sigma - 1)^2(\sigma + 1)$ do not vanish in bands II–III interior). Mirror λ_A : $\mathcal{G}(b, a)$. At most one of $\mathcal{G}(a, b), \mathcal{G}(b, a)$ is positive (their sum is negative on $\sigma \leq 1$... verified numerically on the region — see verification).
8. Mid-dip flush (M_A ; band III, $a > b$): exposed iff the M_A -line lies radially beyond $s_A^{\text{mid}} \cap s_B^{\text{mid}}$; eliminating radicals, the essential factor is

$$(3 + a)R^2 - 3,$$

giving the predicate μ . Mirror: $(3 + b)R^2 - 3$.

9. Corner–vertex identities: the corner between c_A and s_A^{mid} is $V_A^{\text{mid}} = A + u(r_2 - 90^\circ)$, the far endpoint of the side of T_{r_2} through its vertex A (both on c_A , distance 1 from A , and on the side line); similarly $\tilde{V}_A^{\text{mid}} = A + u(r_3 - 90^\circ)$, $V_A^{\text{IV}} = A + u(\varphi_+ - 90^\circ)$, $V_B^{\text{IV}} = B + u(\varphi_- + 90^\circ)$; and the corners between the long arcs and the outer segments are the AB -vertices $W_{\mp}$ of $T_{\mp\psi_1}$ (the unique vertex on both the side through A and the side through B). These are direct consequences of the configuration degenerations: e.g. approaching the end of the admissible interval, the (T2) pivot triangle of E_{BO} converges to T_{φ_e} with x converging to its AB -vertex.

8. Case-by-case: the chains are in U

For each case of `ab_set.md` §4, each listed arc is realizable on exactly the listed parameter range: the constructions of §5 apply, and the endpoint corners are precisely where a containment inequality becomes active (§7 item 9), so consecutive arcs share endpoints. This gives a closed chain $\Gamma \subset U \cap C$, single-valued over $\theta \in [0^\circ, 120^\circ]$ (the corner lists are ordered by strictly increasing θ on the case region — an explicit check of finitely many algebraic inequalities, done in `verify_ab_set.py`). Lemma 5.1 then gives $S \subseteq K$.

9. Case-by-case: maximality ($K \subseteq S$)

Fix a case region and $\theta \in (0^\circ, 120^\circ)$. By Proposition 4.2, $r^*(\theta)$ is the largest radial value among the realizable catalog configurations at direction θ . The claimed chain radius $r_\Gamma(\theta)$ belongs to the catalog, so $r^* \geq r_\Gamma$ is automatic (§8), and it remains to show no other catalog configuration exceeds $r_\Gamma(\theta)$. The finitely many comparisons are of three kinds:

1. **Curve-vs-curve on a sector.** E.g. in case I.6, for θ between $\theta(P_1)$ and $\theta(P_2)$ one checks that L_A dominates c_A , c_B , E_{AO} , E_{BO} , E_{BA} , E_{AB} , the seven h -circles and the other flush lines. This is certified by the following standard semialgebraic scheme.

Lemma 9.1 (sector certification). Let $f, g : (\theta; a, b) \mapsto \mathbb{R}$ be the radial functions of two catalog curves, continuous on a sector bundle $\Sigma = \{(\theta, a, b) : (a, b) \in \Omega, \theta \in [\theta_0(a, b), \theta_1(a, b)]\}$ over a connected case region Ω , with algebraic graphs. If $f = g$ has no solution in the interior of Σ , then $\text{sign}(f - g)$ is constant on the interior of Σ , so a single interior evaluation decides the comparison on all of Σ . The no-solution hypothesis in turn is equivalent to: the intersection locus of the two curves does not meet the open sector for any $(a, b) \in \Omega$ — and every

intersection of two catalog curves is one of the finitely many listed corner points or crossings whose position relative to the sector is controlled by one of the threshold polynomials of §7, which have constant sign on Ω (by the definition of the case regions). $\square$

Each comparison is thus reduced to the corner identities (§7), the constant signs of the threshold polynomials on the case region (definitional), and one interior evaluation per comparison per case region; the interior evaluations are performed at high precision (residual margins $\gg 10^{-6}$, far above the 10^{-10} computational accuracy) in `verify_ab_set.py`.

2. **Non-realizability.** Configurations that are realizable as critical data but whose defining triangle fails to contain all of $\{A, O, B\}$ at direction θ do not enter the max. This prunes: the h -circles inside C (their (S1)/(S2) triangles always lose a containment against the competing (T2)/(T3) configuration that dominates them — e.g. the $|x - O| = h$ circle is dominated by L_A/L_B wherever its configuration is feasible); the AB -flush lines ($\langle x, u(\theta_{AB}) \rangle = d_{AB} - h$ has negative right side, so the line misses C entirely); the variants of (T3b) with third-side support O (their lines pass through regions already dominated by L_A or L_B); and, by Lemma 4.3, the circle $|x - O| = 1$ for $a, b > 0$.
3. **Φ -endpoint capping.** For θ in a dip sector, all interior-critical configurations have already been pruned or dominated, and $\rho_\varphi(\theta)$ is maximized at $\varphi \in \partial\Phi$ (the function $\varphi \mapsto \rho_\varphi(\theta)$ is monotone toward the endpoint on each side of the gap — its derivative is single-signed there because the binding hexagon constraint’s envelope point lies outside the sector); hence $r^*(\theta) = \max(\rho_{\varphi_e^-}, \rho_{\varphi_e^+}, \text{flush lines})$, which is exactly the dip description (segments of T_{φ_e} sides and the exposed flush lines, resolved by $\mathcal{G}, \mu$).

The complete list of comparisons per case, with the verification outcome, is tabulated by `verify_ab_set.py`; every comparison is confirmed on every case region (dense grids, precision 10^{-10} , plus exact sympy checks of all identities quoted in §7). Together with §8 this proves $S = K$ on each case region.

10. Comprehensiveness

The case regions are defined by sign conditions on the finite polynomial family

$$\{ R^2 - 1, 4\sigma^2 - 3, 4R^2 - 3, \sigma - 1, a - \tfrac{1}{2}, b - \tfrac{1}{2}, \sigma - \tfrac{1}{2}, w_+, w_-, \\ \mathcal{E}(a, b), \mathcal{E}(b, a), \mathcal{G}(a, b), \mathcal{G}(b, a), (3 + a)R^2 - 3, (3 + b)R^2 - 3, a - b \}.$$

Every point of $\{a, b \geq 0\}$ satisfies one of: $R^2 > 1$ (Case 0) or the band inequalities I–IV; and within each band the listed sub-cases exhaust all sign patterns that are consistent with the band. The pruning facts, each verified on a fine grid in `verify_ab_set.py` (“region audit”) and provable by elementary estimates:

- $\varepsilon \Rightarrow \alpha \wedge \beta$ (trivially);
- in bands I–III at least one of $w_\pm > 0$ ($w_+ \leq 0 \wedge w_- \leq 0$ forces $\sigma \geq 1$);
- band I $\wedge a \geq \frac{1}{2} \Rightarrow w_+ > 0 \wedge w_- \leq 0$ (so I.6 needs no further split);
- band II excludes ε ($\sigma > \frac{\sqrt{3}}{2} > \frac{1}{2}$);
- within bands II–III never $\mathcal{G}(a, b) > 0 \wedge \mathcal{G}(b, a) > 0$; in band II the flush lines occur only in the rows II.1/II.1’;

- band III $\wedge a > b \Rightarrow a > \frac{1}{2} > b$ (proved in §2-style estimates: $\max(a, b) \leq \frac{1}{2}$ would give $R^2 \leq \frac{3}{4}$, and $a, b > \frac{1}{2}$ would give $\sigma > 1$);
- band III $\wedge a > b$: $(3 + a)R^2 > 3 \Rightarrow \mathcal{G}(a, b) > 0$ (so case “ M_A without λ_B ” is empty);
- band IV admits no side predicates (the corner configurations at P_1, P_2, P'_2, P_3 need $\varphi = 90^\circ$ or 30° admissible, i.e. $F(90^\circ) = \frac{\sqrt{3}}{2}\sigma \leq h$, i.e. $\sigma \leq 1$).

On the common boundaries of adjacent cases both chains degenerate to the same curve list (a vanishing arc), so the closed cases cover the quadrant consistently. ■

Appendix: verified identity list

(All checked exactly in sympy; script `derive_thresholds.py`, `derive_quartics.py`.)

- F -formulas (2.1); $\min F = \frac{\sqrt{3}}{2}R$; band endpoints.
- Implicit quartics of $E_{AO}, E_{BO}, E_{BA}, E_{AB}$ from the parametrizations (6.1).
- $P_1 \in c_A \cap L_A \cap E_{AO}$; $P_2 \in L_A \cap E_{BA} \cap E_{BO}$; $P'_2 \in L_B \cap E_{AB} \cap E_{AO}$; $P_3 \in L_B \cap c_B \cap E_{BO}$.
- Tangency thresholds: $E_{AO}-L_A$ at P_1 : $2a - 1$; $E_{BO}-L_B$ at P_3 : $2b - 1$; $E_{BA}-L_A$ at P_2 : $(b - 1)(2\sigma - 1)$; $E_{AB}-L_B$ at P'_2 : $(a - 1)(2\sigma - 1)$; $E_{BO}-L_A$ at P_2 : $(b - 1)(2b - 1)$; $E_{AO}-L_B$ at P'_2 : $(a - 1)(2a - 1)$.
- Factorizations of the restrictions to L_A, L_B (§7.4) and $\mathcal{C}_B(\frac{1}{2} - a) = -8\mathcal{W}$.
- $P'_2 \in E_{BO} \iff (\sigma - 1)\mathcal{W} = 0$; $P_2 \in E_{AO} \iff (\sigma - 1)(ab^2 - a + b^3 - b^2 + b) = 0$.
- $\mathcal{G}$ (outer flush) and $(3 + a)R^2 - 3$ (mid flush) thresholds by radical elimination.
- Vertex identities of §7.9.